



EC200D&ECx00G&EC600U&EG800G Series

QuecOpen VoLTE API Reference Manual

LTE Standard Module Series

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About the Document

Revision History

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-	2023-08-01	Marvin NING	Creation of the document
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1 Introduction

Quectel EC200D series, ECx00G family, EC600U series, and EG800G-CN modules support QuecOpen[®] solution. QuecOpen[®] is an embedded development platform based on RTOS. It is intended to simplify the design and development of IoT applications. For more information on QuecOpen[®], see **documents [1]** and **[2]**.

This document introduces VoLTE API and the related development example of Quectel EC200D series, ECx00G family, EC600U series, and EG800G-CN modules in QuecOpen[®] solution.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC200D Series
ECx00G	EC200G-CN
	EC600G-CN
	EC800G-CN
	EC600U Series
-	EG800G-CN

NOTE

1. For the EC200D series module, the file paths mentioned in this document do not contain *components*.
2. To support VoLTE feature, you need to add VoLTE when compiling the software versions of ECx00G family and EC600U series modules, for example: **build_all new EC600UCN_LB EC600UCNLB VoLTE.**

2 VoLTE API

2.1. Header File

ql_api_volte.h, the header file of VoLTE API, is in the *components\ql-kernel\inc* directory. Unless otherwise specified, all the header files mentioned in this document are in this directory.

2.2. API Overview

Table 2: API Overview

Function	Description
<i>ql_volte_set()</i>	Sets VoLTE feature.
<i>ql_volte_get()</i>	Gets the status of VoLTE feature.
<i>ql_volte_vdp_set()</i>	Sets voice domain preference.
<i>ql_volte_vdp_get()</i>	Gets voice domain preference.
<i>ql_volte_usage_set()</i>	Sets VoLTE usage of the module.
<i>ql_volte_usage_get()</i>	Gets VoLTE usage of the module.

2.3. API Description

2.3.1. ql_volte_set

This function sets VoLTE feature.

- **Prototype**

```
ql_volte_errcode_e ql_volte_set(uint8_t nSim,uint8_t mode);
```

- **Parameter**

nSim:

[In] (U)SIM card in use. Set this parameter to 0 if the module supports only 1 (U)SIM interface.

0 (U)SIM 1
1 (U)SIM 2

mode:

[In] Enable or disable VoLTE feature. Default: 1.

0 Disable
1 Enable

- **Return Value**

See **Chapter 2.3.1.1** for details.

NOTE

When VoLTE feature is enabled, the range of *profile_idx* in *ql_start_data_call()* is 1–5 and manual activation of PDP context 6 or 7 is not allowed. See **document [3]** for more information about *ql_start_data_call()*.

2.3.1.1. ql_volte_errcode_e

The enumeration of VoLTE API result codes:

```
typedef enum
{
    QL_VOLTE_SUCCESS                = 0,
    QL_VOLTE_ERROR                  = 1 | (QL_COMPONENT_IMS << 16),
    QL_VOLTE_PARA_ERR,
}ql_volte_errcode_e;
```

- Member

Member	Description
QL_VOLTE_SUCCESS	Successful execution
QL_VOLTE_ERROR	Failed execution
QL_VOLTE_PARA_ERR	Parameter error

2.3.2. ql_volte_get

This function gets the status of VoLTE feature.

- Prototype

```
ql_volte_errcode_e ql_volte_get(uint8_t nSim, uint8_t *mode);
```

- Parameter

nSim:

[In] (U)SIM card in use.

0 (U)SIM 1

1 (U)SIM 2

mode:

[Out] Pointer to the status of VoLTE feature.

0 VoLTE disabled

1 VoLTE enabled

- Return Value

See **Chapter 2.3.1.1** for details.

2.3.3. ql_volte_vdp_set

This function sets voice domain preference.

- Prototype

```
ql_volte_errcode_e ql_volte_vdp_set( uint8_t nSim, uint8_t setting);
```

● Parameter

nSim:

[In] (U)SIM card in use. Set this parameter to 0 if the module supports only 1 (U)SIM interface.

- 0 (U)SIM 1
- 1 (U)SIM 2

setting:

[In] Voice domain preference. Default: 3.

- 1 CS Voice only
- 2 CS Voice preferred, IMS PS Voice as secondary
- 3 IMS PS Voice preferred, CS Voice as secondary
- 4 IMS PS Voice only

● Return Value

See **Chapter 2.3.1.1** for details.

2.3.4. ql_volte_vdp_get

This function gets voice domain preference.

● Prototype

```
ql_volte_errcode_e ql_volte_vdp_get(uint8_t nSim, uint8_t *setting);
```

● Parameter

nSim:

[In] (U)SIM card in use.

- 0 (U)SIM 1
- 1 (U)SIM 2

setting:

[Out] Pointer to voice domain preference.

- 1 CS Voice only
- 2 CS Voice preferred, IMS PS Voice as secondary
- 3 IMS PS Voice preferred, and CS Voice as secondary
- 4 IMS PS Voice only

● Return Value

See **Chapter 2.3.1.1** for details.

2.3.5. ql_volte_usage_set

This function sets VoLTE usage of the module.

● Prototype

```
ql_volte_errcode_e ql_volte_usage_set(uint8_t nSim, uint8_t setting);
```

● Parameter

nSim:

[In] (U)SIM card in use. Set this parameter to 0 if the module supports only 1 (U)SIM interface.

- 0 (U)SIM 1
- 1 (U)SIM 2

setting:

[In] VoLTE usage setting. Default: 1.

- 0 Voice centric
- 1 Data centric

● Return Value

See **Chapter 2.3.1.1** for details.

2.3.6. ql_volte_usage_get

This function gets VoLTE usage of the module.

● Prototype

```
ql_volte_errcode_e ql_volte_usage_get(uint8_t nSim, uint8_t *setting);
```

● Parameter

nSim:

[In] (U)SIM card in use.

- 0 (U)SIM 1
- 1 (U)SIM 2

setting:

[Out] Pointer to VoLTE usage setting.

- 0 Voice centric
- 1 Data centric

● Return Value

See **Chapter 2.3.1.1** for details.

3 Development Example

This chapter introduces how to use the above API to develop and debug VoLTE on the application layer.

3.1. Development Example

volte_demo.c, the example file of VoLTE API, provided in the QuecOpen SDK of the module, is in the `\components\ql-application\volte` directory. The entry function is *ql_volte_app_init()* which initializes *volte_demo_task*, as shown below.

```
QlOssStatus ql_volte_app_init(void)
{
    QlOssStatus err = QL_OSI_SUCCESS;

    err = ql_rtos_task_create(&volte_task, 2048, APP_PRIORITY_NORMAL, "volte_task", volte_demo_task, NULL, 2);
    if(err != QL_OSI_SUCCESS)
    {
        QL_VOLTE_LOG("volte_demo_task created failed, ret = 0x%x", err);
    }

    return err;
}
```

As shown in the figure below, *ql_volte_app_init()* is disabled by default; if necessary, uncomment it in *components\quectel\ql_open\ql_examples\init\ql_init.c* file.

```
#ifdef QL_APP_FEATURE_VOLTE
    ql_volte_app_init();
#endif
```

3.2. Feature Debugging

The module's VoLTE feature can be debugged on Quectel LTE OPEN EVB.

Download the compiled firmware to the module and connect the USB port of the LTE OPEN EVB to a PC with a USB cable. USB AP Log Port and USB DIAG Port mainly display the system debugging information. You can view information of the example through USB AP Log Port or USB DIAG Port after capturing the log with cooltools or ArmLogel. For details about log capture, see **documents [4], [5] and [6]**.

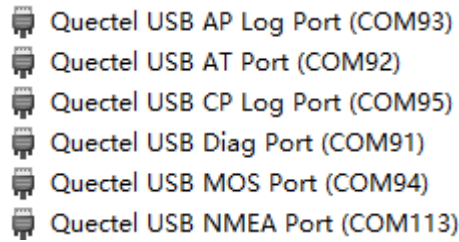


Figure 1: Ports in Device Manager (EC600U Series)

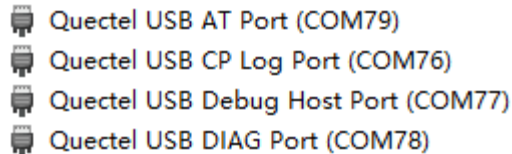


Figure 2: Ports in Device Manager (EC200D Series/ECx00G Family/EG800G-CN)

NOTE

For EC600U series module, cooltools is used to capture logs through USB AP Log Port; see [document \[4\]](#) for details on log capture. For EC200D series, ECx00G family and EG800G-CN modules, ArmLogel is used to capture logs through USB DIAG Port; see [documents \[5\]](#) and [\[6\]](#) for details on log capture.

After *volte_demo_task* starts, it automatically enables VoLTE feature and sets data centric and IMS PS Voice preferred, as shown below.

```
void volte_demo_task(void * param)
{
    ql_volte_errcode_e err = QL_VOLTE_SUCCESS;
    uint8_t nSim = 0;

    //China Telecom 4G/5G SIM card must enable VoLTE before making a call
    err = ql_volte_set(nSim, 1);
    if(err != QL_VOLTE_SUCCESS){
        QL_VOLTE_LOG("ql_voice_call_volte_set FAIL");
    }else {
        QL_VOLTE_LOG("ql_voice_call_volte_set OK");
    }

    //data centric
    err = ql_volte_usage_set(nSim, 1);
    if(err != QL_VOLTE_SUCCESS){
        QL_VOLTE_LOG("data centric set FAIL,err = %d",err);
    }else{
        QL_VOLTE_LOG("data centric set OK");
    }

    //set IMS PS Voice preferred, CS Voice as secondary
    err = ql_volte_vdp_set(nSim, 4);
    if(err != QL_VOLTE_SUCCESS){
        QL_VOLTE_LOG("set IMS PS Voice preferred FAIL,err=%d",err);
    }else{
        QL_VOLTE_LOG("set IMS PS Voice preferred OK");
    }

    ql_rtos_task_delete(NULL);
} « end volte_demo_task »
```

4 Appendix References

Table 3: Related Documents

Document Name
[1] Quectel_ECx00G&EC600U&EG800G_Series_QuecOpen_CSDK_Quick_Start_Guide
[2] Quectel_EC200D_Series_QuecOpen_CSDK_Quick_Start_Guide
[3] Quectel_EC200D&ECx00G&EC600U&EG800G_Series_QuecOpen_Data_Call_API_Reference_Manual
[4] Quectel_EC600U_Series_QuecOpen_Log_Capture_Guide
[5] Quectel_EC200D_Series_Log_Capture_Guide
[6] Quectel_ECx00G&EG800G_Series_QuecOpen_Log_Capture_Guide

Table 4: Terms and Abbreviations

Abbreviation	Description
(U)SIM	(Universal) Subscriber Identity Module
API	Application Programming Interface
CS	Circuit Switching
EVB	Evaluation Board
IMS	IP Multimedia Subsystem
IoT	Internet of Things
LTE	Long Term Evolution
PC	Personal Computer
PS	Packet Switch

RTOS	Real-Time Operating System
SDK	Software Development Kit
USB	Universal Serial Bus
VoLTE	Voice (voice calls) over LTE